



AMC ENGINEERING COLLEGE
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Capacity Planning for IT

Sub Code:BCS456B

MODULE-4

1. What is the procurement pipeline?

- A. The process by which we obtain new capacity
- B. The process of developing new software
- C. The process of hiring new employees
- D. The process of marketing new products

Answer: A. The process by which we obtain new capacity

2. What is a significant challenge in the procurement process for large organizations?

- A. Quick delivery times
- B. Long approval times for purchasing hardware
- C. Lack of developers
- D. Insufficient marketing strategies

Answer: B. Long approval times for purchasing hardware

3. What metric is critical to keep an eye on when ordering equipment?

- A. CPU usage
- B. Memory allocation
- C. Procurement time
- D. Network speed

Answer: C. Procurement time

4. What is Just-in-Time (JIT) inventory practice?

- A. Maintaining large inventories to ensure availability
- B. Ordering parts and stocking them only as they are needed
- C. Increasing storage capacity before it is needed
- D. Reducing the number of suppliers

Answer: B. Ordering parts and stocking them only as they are needed

5. Why might startups order servers out of fear?

- A. To reduce operational costs
- B. To ensure they have enough capacity when launching new products
- C. To improve code quality
- D. To optimize marketing efforts

Answer: B. To ensure they have enough capacity when launching new products

6. Which step in the procurement process involves creating graphs with attention-grabbing fonts?

- A. Determine the needs
- B. Justify purchases
- C. Solicit quotes from vendors
- D. Testing

Answer: B. Justify purchases

7. What is one benefit of using memcached in web applications?

- A. Increasing database query times
- B. Reducing load on database servers
- C. Increasing storage costs

D. Slowing down the response time to clients

Answer: B. Reducing load on database servers

8. What can long-term trends in traffic patterns help inform?

A. Immediate hardware needs

B. Long-term decisions and site strategy

C. Short-term marketing efforts

D. Daily operational tasks

Answer: B. Long-term decisions and site strategy

9. What is an example of a typical daily traffic pattern for a web server in the United States?

A. Load rises slowly in the morning, peaks in the afternoon, and drops overnight

B. Load remains constant throughout the day

C. Load drops in the morning and rises in the evening

D. Load peaks overnight and drops during the day

Answer: A. Load rises slowly in the morning, peaks in the afternoon, and drops overnight

10. What could be a side effect of deploying memcached?

A. Decreased web server load

B. Increased web server capacity issues

C. Slower user response times

D. Reduced database query speeds

Answer: B. Increased web server capacity issues

11. Why is understanding user behavior important in capacity planning?

A. It helps determine the company's financial health

B. It provides valuable data for predicting future demand

C. It ensures better marketing strategies

D. It reduces the need for technical support

Answer: B. It provides valuable data for predicting future demand

12. Which metric would you track to understand user engagement in an online community?

A. CPU usage

B. Posts per day, per user

C. Disk I/O

D. Network speed

Answer: B. Posts per day, per user

13. What is the relationship between capacity planning and product management?

A. They are completely independent of each other

B. Product management develops ideas, while capacity planning focuses on maintaining current infrastructure

C. Both use data for forecasting, and their efforts should be aligned

D. Capacity planning only relies on historical data, not product plans

Answer: C. Both use data for forecasting, and their efforts should be aligned

14. What is a best practice for capacity planners in relation to product management?

A. Isolate from product management to avoid interference

B. Develop an ongoing conversation with product management

C. Rely solely on historical data for planning

D. Avoid discussing new features with product management

Answer: B. Develop an ongoing conversation with product management

15. How often should forecasts be reviewed for accuracy?

- A. Monthly
- B. Bi-annually
- C. Weekly or even daily
- D. Annually

Answer: C. Weekly or even daily

16. In the provided example, how accurate was the prediction regarding storage consumption?

- A. Exact
- B. More time than expected was available
- C. Less time than expected was available
- D. Completely inaccurate

Answer: B. More time than expected was available

17. What should be done if hardware limitations change?

- A. Continue using the old ceilings
- B. Update the metric used to determine the ceiling
- C. Ignore the changes until the next review
- D. Focus only on software improvements

Answer: B. Update the metric used to determine the ceiling

18. What is a possible seasonal variation mentioned in the text?

- A. Increased usage during the summer holidays
- B. Decreased usage during weekdays
- C. Increased usage during the start of college in the fall
- D. Decreased usage during the holiday season in November and December

Answer: C. Increased usage during the start of college in the fall

19. What effect do seasonal variations have on forecasting?

- A. They simplify the forecasting process
- B. They make forecasting unnecessary
- C. They influence how wide or narrow the forecasting window should be
- D. They reduce the need for hardware upgrades

Answer: C. They influence how wide or narrow the forecasting window should be

20. What can help identify flawed assumptions in capacity planning?

- A. Ignoring historical data
- B. Consistently reexamining trends and questioning past predictions
- C. Isolating from product management
- D. Using only one metric for all planning

Answer: B. Consistently reexamining trends and questioning past predictions

21. What is one of the key goals of automated deployment and configuration tools?

- A. Increase time to provision new capacity
- B. Minimize time to provision new capacity
- C. Increase hardware diversity
- D. Decentralize configuration management

Answer: B. Minimize time to provision new capacity

22. What advantage does a centralized configuration and management environment provide?

- A. Increases the complexity of the infrastructure
- B. Makes replication and backup of files easier to manage
- C. Eliminates the need for version control
- D. Allows multiple servers to perform configuration tasks

Answer: B. Makes replication and backup of files easier to manage

23. Why should logging into an individual server be restricted?

- A. To prevent unauthorized access
- B. To ensure all management is done from a central location
- C. To make it harder to troubleshoot
- D. To allow different configuration settings for different hardware

Answer: B. To ensure all management is done from a central location

24. What is the “holy grail” of all deployment systems according to the text?

- A. Have administrators manually install software on new servers
- B. Have new servers automatically start working without administrator intervention
- C. Keep hardware inventory in a distributed manner
- D. Perform all configuration changes directly on individual servers

Answer: B. Have new servers automatically start working without administrator intervention

25. How does maintaining consistency across a cluster of servers help in troubleshooting?

- A. It makes it harder to identify bugs
- B. It ensures that software configurations are different
- C. It eliminates configuration inconsistencies as a source of problems
- D. It increases the number of changes to track

Answer: C. It eliminates configuration inconsistencies as a source of problems

26. What is a potential hindrance of having disparate software configurations across nodes in a given cluster?

- A. It makes the system easier to optimize
- B. It limits performance optimization

- C. It simplifies the troubleshooting process
- D. It eliminates the need for a centralized management environment

Answer: B. It limits performance optimization

27. What is the benefit of having version control in a centralized configuration and management environment?

- A. It increases the number of repeated tasks
- B. It makes changes immediately apparent
- C. It hides the changes from multiple users
- D. It complicates the tracking of changes

Answer: B. It makes changes immediately apparent

28. Why is it important to coordinate troubleshooting efforts during outages or emergencies?

- A. To ensure each server has a different OS
- B. To prevent the infrastructure from becoming homogeneous
- C. To effectively enable coordination and eliminate inconsistencies
- D. To allow for more configuration discrepancies

Answer: C. To effectively enable coordination and eliminate inconsistencies

29. What is one advantage of package-based systems for OS installation?

- A. Faster installation process
- B. Easier to see what's installed due to package management tools
- C. Uses fewer system resources
- D. Allows for direct filesystem manipulation

Answer: B. Easier to see what's installed due to package management tools

29. What is one advantage of image-based systems for OS installation?

- A. Greater accountability for installed files
- B. Faster installation process

- C. Uses package management tools
- D. More complex to manage

Answer: B. Faster installation process

30. What does an Amazon Machine Image (AMI) include?

- A. Only the OS
- B. Only application servers
- C. A template for the root volume, launch permissions, and block device mapping
- D. A template for the root volume and application servers only

Answer: C. A template for the root volume, launch permissions, and block device mapping

31. How do configuration management systems help with configuration tasks?

- A. By eliminating the need for backups
- B. By organizing configuration files into useful subsystems
- C. By requiring manual updates for each machine
- D. By hiding institutional knowledge

Answer: B. By organizing configuration files into useful subsystems

32. What is the role of a client process in a typical configuration management system?

- A. To manually request configuration updates
- B. To periodically poll the server for new configurations
- C. To replace the server in case of failure
- D. To store all configurations locally

Answer: B. To periodically poll the server for new configurations

33. In the context of configuration management, what is a "role"?

- A. A particular type of machine that performs a specific task
- B. A collection of unrelated software and configurations
- C. A single server dedicated to a unique task
- D. A temporary setting used for testing purposes

Answer: A. A particular type of machine that performs a specific task

34. What is an example of a service in a web server role?

- A. Only the httpd package
- B. The httpd package, its configuration, and related metrics and health checks
- C. Any unrelated software running on the server
- D. A single setting for any metric

Answer: B. The httpd package, its configuration, and related metrics and health checks

35. Why is it beneficial to identify services common to multiple roles?

- A. To create redundant configuration routines
- B. To simplify deploying new roles by reusing existing configurations
- C. To ensure each role is unique and independent
- D. To avoid using any common configurations

Answer: B. To simplify deploying new roles by reusing existing configurations